

CSE/ECE 511 Computer Architecture: Quiz 1 Rubric
Set A

Ans 1.

Let

N = total number of instructions

Na = number of type I instructions

Nb = number of type II instructions

F = frequency of the CPUs

Therefore,

$$\text{Execution time for CPUs} = [(Na * 8) + (Nb * 32)] * 1/F$$

[2 marks for equation]

Where

For CPU-A, $Na = Nb = N/2$

For CPU-B, $Na = xN$ and $Nb = (1-x)N$ where x is in $[0, 1]$

Therefore

$$20 = [(N/2 * 8) + (N/2 * 32)] * 1/F$$

[1 mark]

$$10 = [(Nx * 8) + (N(1-x) * 32)] * 1/F$$

[2 marks]

$$\Rightarrow [(N/2 * 8) + (N/2 * 32)] = 2 * [(Nx * 8) + (N(1-x) * 32)]$$

[2 marks]

$$\Rightarrow (4N + 16N) = 2 * [8Nx + 32N(1-x)]$$

$$\Rightarrow 20N = 2N[8x + 32 - 32x]$$

$$\Rightarrow 10 = 8x + 32 - 32x$$

$$\Rightarrow 10 = 32 - 24x$$

$$\Rightarrow 24x = 22$$

$$\Rightarrow x = 11/12 = 0.92$$

[1 mark]

Na for CMP-B = 0.92N

Nb for CMP-B = 0.08N

Therefore a 92%-8% split.

[2 marks]

Note: Ratio , without %age conversion should also be awarded full marks.

Ans2.

Expected latency

$$= (\text{hit_rate}) * (\text{hit_check_latency} + \text{access_latency}) + (1 - \text{hit_rate}) * (\text{hit_check_latency} + \text{expected_latency_of_next_level})$$

$$\begin{aligned} \text{For RAM, expected latency} &= 1 * (0 + 60,000) + 0 \\ &= 60,000 \end{aligned}$$

[1 mark]

$$\begin{aligned} \text{For L3, expected latency} &= 0.9 * (5,000 + 5,000) + 0.1 * (5,000 + 60,000) \\ &\quad [1 \text{ mark for correct hit} + 1 \text{ mark for correct miss}] \\ &= (0.9 * 10,000) + (0.1 * 65,000) \end{aligned}$$

$$= 9,000 + 6,500$$

$$= 15,500 \quad [1 \text{ mark}]$$

For L2, expected latency

$$= 0.8 * (500 + 500) + 0.2 * (500 + 15,500)$$

[1 mark for correct hit + 1 mark for correct miss]

$$= (0.8 * 1000) + (0.2 * 16,000)$$

$$= 800 + 3,200$$

$$= 4,000 \quad [1 \text{ mark}]$$

For L1, expected latency

$$= 0.7 * (50 + 50) + 0.3 * (50 + 4,000)$$

[1 mark for correct hit + 1 mark for correct miss]

$$= (0.7 * 100) + (0.3 * 4,050)$$

$$= 70 + 1,215$$

$$= 1,285 \quad [1 \text{ mark}]$$

Set B

Ans 1.

Let

N = total number of instructions

Na = number of type I instructions

Nb = number of type II instructions

F = frequency of the CPUs

Therefore,

$$\text{Execution time for CPUs} = [(Na * 8) + (Nb * 32)] * 1/F$$

[2 marks for equation]

Where

For CPU-A, Na = Nb = N/2

For CPU-B, Na = xN and Nb = (1-x)N where x is in [0,1]

Therefore

$$20 = [(N/2 * 8) + (N/2 * 32)] * 1/F \quad [1 \text{ mark}]$$

$$16 = [(Nx * 8) + (N(1-x) * 32)] * 1/F \quad [2 \text{ marks}]$$

$$\Rightarrow [(N/2 * 8) + (N/2 * 32)] = (5/4) * [(Nx * 8) + (N(1-x) * 32)] \quad [2 \text{ marks}]$$

$$\Rightarrow (4N + 16N) = (5/4) * [8Nx + 32N(1-x)]$$

$$\Rightarrow 20N = (5/4) * N * [8x + 32 - 32x]$$

$$\Rightarrow 16 = 8x + 32 - 32x$$

$$\Rightarrow 16 = 32 - 24x$$

$$\Rightarrow 24x = 16$$

$$\Rightarrow x = 2/3 = 0.667 \quad [1 \text{ mark}]$$

Na for CMP-B = 0.66N

Nb for CMP-B = 0.333N

Therefore a 66.7%-33.3% split.

[2 marks]

Ans2.

Expected latency

$$= (\text{hit_rate}) * (\text{hit_check_latency} + \text{access_latency}) + (1 - \text{hit_rate}) * (\text{hit_check_latency} + \text{expected_latency_of_next_level})$$

$$\begin{aligned} \text{For RAM, expected latency} &= 1 * (0 + 60,000) + 0 \\ &= 60,000 \end{aligned}$$

[1 mark]

$$\begin{aligned} \text{For L3, expected latency} &= 0.95 * (5,000 + 5,000) + 0.05 * (5,000 + 60,000) \\ &\quad [1 \text{ mark for correct hit} + 1 \text{ mark for correct miss}] \\ &= (0.95 * 10,000) + (0.05 * 65,000) \\ &= 9,500 + 3,250 \\ &= 12,750 \end{aligned}$$

[1 mark]

$$\begin{aligned} \text{For L2, expected latency} &= 0.85 * (500 + 500) + 0.15 * (500 + 12,750) \\ &\quad [1 \text{ mark for correct hit} + 1 \text{ mark for correct miss}] \\ &= (0.85 * 1000) + (0.15 * 13,250) \\ &= 850 + 1,987.5 \\ &= 2837.5 \end{aligned}$$

[1 mark]

$$\begin{aligned} \text{For L1, expected latency} &= 0.75 * (50 + 50) + 0.25 * (50 + 2837.5) \\ &\quad [1 \text{ mark for correct hit} + 1 \text{ mark for correct miss}] \\ &= (0.75 * 100) + (0.25 * 2887.5) \\ &= 75 + 721.875 \\ &= 796.875 \end{aligned}$$

[1 mark]

Set C

Ans 1.

Let

N = total number of instructions

Na = number of type I instructions

Nb = number of type II instructions

F = frequency of the CPUs

Therefore,

$$\text{Execution time for CPUs} = [(Na * 8) + (Nb * 32)] * 1/F$$

[2 marks for equation]

Where

For CPU-A, Na = Nb = N/2

For CPU-B, Na = xN and Nb = (1-x)N where x is in [0,1]

Therefore

$$20 = [(N/2 * 8) + (N/2 * 32)] * 1/F \quad [1 \text{ mark}]$$

$$24 = [(Nx * 8) + (N(1-x) * 32)] * 1/F \quad [2 \text{ marks}]$$

$$\frac{20}{24} = \frac{4N + 16N}{8Nx + 32N(1-x)}$$

$$\frac{5}{6} = \frac{20}{-24x + 32}$$

$$-120x + 160 = 120$$

$$x = \frac{1}{3}$$

$$N_a \text{ for CMP-B} = N/3$$

$$N_b \text{ for CMP-B} = 2N/3$$

Therefore a 1:2 or 33.3% & 66.7% split. [2 marks]

Ans2.

Expected latency

$$= (\text{hit_rate}) * (\text{hit_check_latency} + \text{access_latency}) + (1 - \text{hit_rate}) * (\text{hit_check_latency} + \text{expected_latency_of_next_level})$$

$$\begin{aligned} \text{For RAM, expected latency} &= 1 * (0 + 60,000) + 0 \\ &= 60,000 \end{aligned} \quad [1 \text{ mark}]$$

$$\begin{aligned} \text{For L3, expected latency} &= 0.85 * (5,000 + 5,000) + 0.15 * (5,000 + 60,000) \\ &\quad [1 \text{ mark for correct hit} + 1 \text{ mark for correct miss}] \\ &= 8500 + 9750 \\ &= 18,250 \end{aligned} \quad [1 \text{ mark}]$$

$$\begin{aligned} \text{For L2, expected latency} &= 0.75 * (500 + 500) + 0.25 * (500 + 18,250) \\ &\quad [1 \text{ mark for correct hit} + 1 \text{ mark for correct miss}] \\ &= 750 + 4687.5 \\ &= 5437.5 \end{aligned} \quad [1 \text{ mark}]$$

$$\begin{aligned} \text{For L1, expected latency} &= 0.65 * (50 + 50) + 0.35 * (50 + 5437.5) \\ &\quad [1 \text{ mark for correct hit} + 1 \text{ mark for correct miss}] \\ &= 65 + 1920.625 \\ &= 1,985.625 \end{aligned} \quad [1 \text{ mark}]$$